



Perimeter, Area & Algebraic Expressions

Day 16 of 20 · Duration: 50 minutes · Theme: Generic Amusement Park

MINNESOTA STANDARDS (MDE 2022)

Standard: 9.3.5.7

Use the structure of an expression, equation and/or formula to create an equivalent form that is more helpful given the situation. Rearrange formulas to highlight a quantity of interest, using the same reasoning in solving equations. (MP6)

Standard: 9.3.5.8

Use the structure of an expression to write it in multiple ways.

ESSENTIAL QUESTIONS

- How do we write algebraic expressions for perimeter and area?
- How does a change in a side length change the perimeter?
- Why does the same formula work for different shapes of the same type?

VOCABULARY

Perimeter — The distance around a shape. **Area** — The amount of space inside a shape.
Dimension — A measurable side of a shape: length, width, or height.

LEARNING OBJECTIVES & SUCCESS CRITERIA

- I can write and evaluate algebraic expressions for perimeter and area.

- I can show my work step by step and check my answers for reasonableness.
- I can explain my reasoning to a partner using math vocabulary.

ENGAGEMENT STRATEGY

Strategy: Collaborative Whiteboard

Practice Format: Large Group Practice

Expectation: Every student responds on a whiteboard or to a partner before any answer is shared.

BELL RINGER — WHITEBOARDS READY!

Prompt: Write an expression for the perimeter of a square with side length s . *Students write their response on a whiteboard and hold it up. Teacher scans for misconceptions before moving on.*

LESSON PACING (50 MINUTES)

Phase 1: Bell Ringer 5 min min

Write an expression for the perimeter of a square with side length s .

Phase 2: Direct Instruction 10 min min

Perimeter, Area & Algebraic Expressions — teacher-led notes (I Do); GEMDAS reference as needed.

Phase 3: Guided Practice 12 min min

Work together (We Do) — Large Group Practice. Solve together: A rectangle's length is 3 more than its width w . Write an expression for the perimeter. -> Answer: $4w + 6$.

Phase 4: Culture Connection 8 min min

Sami (Sápmi / Northern Europe) — The Sámi are the Indigenous people of Sápmi, spanning northern Norway, Sweden, Finland, and (with math word problem)

Phase 5: Practice & Game 10 min min

GO: A rectangle's length is 3 more than its width w . Write an expression for the perimeter. | SET: A square has side s . Write expressions for area and perimeter. A triangle has sides x , $2x$, and $x + 5$. Write its perimeter. | EXTRA: A rectangle has area 48 and width 6. Write and solve an equation for the length l .

Phase 6: Exit Ticket 5 min min

A rectangle has length $(x + 4)$ and width 3. Write an expression for its perimeter.

PRACTICE PROBLEMS (GO / SET / EXTRA)

GO: A rectangle's length is 3 more than its width w . Write an expression for the perimeter. **Answer:** $4w + 6$

SET: A square has side s . Write expressions for area and perimeter. A triangle has sides x , $2x$, and $x + 5$. Write its perimeter. **Answer:** $A = s^2$ squared, $P = 4s$; $4x + 5$

EXTRA: A rectangle has area 48 and width 6. Write and solve an equation for the length l . **Answer:** $6l = 48, l = 8$

CULTURE CONNECTION — MINNESOTA STORIES

Spotlight: Sami (Sápmi / Northern Europe)

The Sámi are the Indigenous people of Sápmi, spanning northern Norway, Sweden, Finland, and Russia. Sami culture centers on reindeer herding and duodji (handicraft).

Math Word Problem: A Sámi garment maker decorates a rectangular luhkka (reindeer herder) cloak. The length is $3x + 2$ and the width is $2x - 1$, all in units. Write an expression for the perimeter of the rectangle.

CAREER SPOTLIGHT — MINNESOTA

Career: Andersen Corp — Bayport | Window Energy Rater | AA | \$80,142 MN DEED 25th-percentile wage; OES-cited. Connection: algebra skills used daily on the job.

DIFFERENTIATION

- Scaffolding: sentence starters for translation tasks; GEMDAS reference card on desks.
- Extension: students write and solve their own word problems using the lesson's operation keywords.

- Language support: bilingual vocabulary cards; pair students with a math-language peer.
- Grouping: flexible pairs/trios by readiness; Merit Time games rotate roles each round.

ASSESSMENT

- Formative: whiteboard checks during the bell ringer; circulation during GO/SET/EXTRA; exit ticket each lesson.
- Game-based: Merit Time games give instant fluency feedback (L1–L5).
- Summative: Quiz 1 (Day 7, L1–L5), Quiz 2 (Day 17, L6–L15), End-of-Unit Test (Day 20, L1–L17).